Representation Learning with Contrastive Predictive Coding (InfoNCE) (2018; v2 2019)

Authors: Aaron van den Oord, Yazhe Li, Oriol Vinyals (DeepMind)
PDF: `https://arxiv.org/abs/1807.03748`

Challenges

- Learn representations from high-dimensional sequential data without reconstructing pixels/waveforms or relying on labels.
- Avoid trivial prediction objectives (too easy) or overly hard ones; create a task that forces extracting shared/high-level information over time.
- Scale contrastive learning stably with many negatives and across domains (speech, vision, NLP, RL).

Key Initiatives

- Propose CPC: encode observations into latents, summarize past with an autoregressive context, then predict future latents via a probabilistic contrastive objective.
- Introduce/coin **InfoNCE**: an NCE-style loss implemented as *N-way classification* among one positive and many negatives.
- Show InfoNCE provides a mutual-information lower bound: *"*$I(x_{t+k}, c_t) \geq \log(N) - L_N$*"* (Sec. 2.3).

Methodology

- **Architecture:** encoder g_{enc} maps each input $x_t \to z_t$; autoregressive model g_{ar} summarizes history into context c_t; prediction heads score (c_t, x_{t+k}) pairs.
- **InfoNCE definition (Eq. 4):** given a set X with 1 positive from $p(x_{t+k} \mid c_t)$ and $N-1$ negatives from the proposal $p(x_{t+k})$, optimize

$$L_N = -\mathbb{E} \log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)}.$$

- The paper explicitly interprets Eq. 4 as *"the categorical cross-entropy of classifying the positive sample correctly"* (Sec. 2.3), and shows the optimal scorer estimates a **density ratio** (Eq. 5).

InfoNCE vs other losses (key differences)

- **Vs. classic Noise-Contrastive Estimation (NCE):** CPC frames learning as NCE-style discrimination (Sec. 2.3), but InfoNCE is implemented as **multi-class (N-way) cross-entropy** over a set with 1 positive + many negatives (Eq. 4), rather than independent binary decisions. InfoNCE is also used to derive an **MI lower bound** (Sec. 2.3 / App. A.1).
- **Vs. triplet / max-margin ranking losses:** the paper cites prior contrastive work using *"triplet losses using a max-margin approach to separate positive from negative examples"* (Sec. 2.4). Compared to margin ranking (distance + hinge), InfoNCE is **probabilistic** (softmax normalization across all negatives), giving smoother gradients and a direct interpretation as classification among candidates.
- **Vs. supervised cross-entropy classification:** InfoNCE uses the same mathematical form (categorical cross-entropy), but the "label" is **which element in the candidate set is the positive sample** (self-supervised), not a fixed semantic class.

- **Vs. maximum-likelihood generative prediction (sequence decoders):** the paper contrasts CPC's latent predictive objective with language models that **"use maximum likelihood over sequences of observations"** (Sec. 2.4). InfoNCE avoids reconstructing raw inputs and instead learns representations by predicting **latent future information** that is shared with context.
- **Vs. MINE / DV-style MI estimators:** the appendix relates InfoNCE to MINE (Belghazi et al., 2018) and notes direct MINE can be **unstable when the target is easy**, whereas InfoNCE remains stable: *"[MINE] became very unstable if the target was easy to predict from the context"* (App. A.1).

Experiments (selected)

- **Speech (LibriSpeech):** phone classification accuracy improves from 39.7 (MFCC) to 64.6 (CPC); speaker classification reaches 97.4 (Table 1).
- **Vision (ImageNet linear eval):** CPC reports 48.7 top-1 and 73.6 top-5 (Tables 3–4), with the text highlighting *"9% absolute"* top-1 gain over prior state of the art.
- **NLP (BookCorpus transfer):** comparable to Skip-thought on several tasks and strong on TREC (96.8) (Table 5).
- **RL (DeepMind Lab):** CPC used as an auxiliary loss improves learning curves across multiple tasks (Fig. 6).

References (selected; cited by the paper)

- Noise-Contrastive Estimation (Gutmann & Hyvärinen, 2010; Mnih & Teh, 2012; Mnih & Kavukcuoglu, 2013) [12,14,15]
- Word2Vec negative sampling (Mikolov et al., 2013) [9]
- Triplet / max-margin contrastive learning (e.g., [21,22,23])
- PixelCNN / masked conv autoregressive modeling (van den Oord et al., 2016) [19]
- MINE: Mutual Information Neural Estimation (Belghazi et al., 2018) [54]

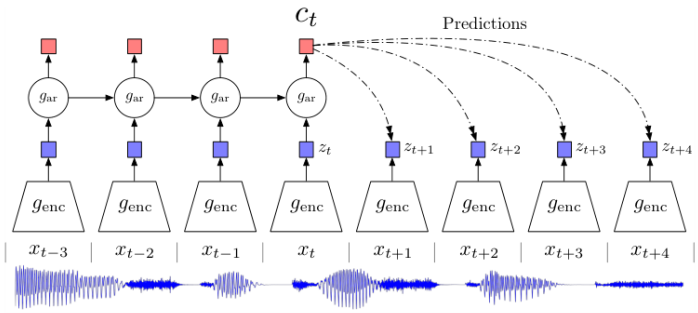

Figure 1: Overview of Contrastive Predictive Coding, the proposed representation learning approach. Although this figure shows audio as input, we use the same setup for images, text and reinforcement learning.

shown useful [10, 11]. We hypothesize that these approaches are fruitful partly because the context from which we predict related values are often conditionally dependent on the same shared high-level latent information. And by casting this as a prediction problem, we automatically infer these features of interest to representation learning.

In this paper we propose the following: first, we compress high-dimensional data into a much more compact latent embedding space in which conditional predictions are easier to model. Secondly, we use powerful autoregressive models in this latent space to make predictions many steps in the future. Finally, we rely on Noise-Contrastive Estimation [12] for the loss function in similar ways that have been used for learning word embeddings in natural language models, allowing for the whole model to be trained end-to-end. We apply the resulting model, Contrastive Predictive Coding (CPC) to widely different data modalities, images, speech, natural language and reinforcement learning, and show that the same mechanism learns interesting high-level information on each of these domains, outperforming other approaches.

2 Contrastive Predicting Coding

We start this section by motivating and giving intuitions behind our approach. Next, we introduce the architecture of Contrastive Predictive Coding (CPC). After that we explain the loss function that is based on Noise-Contrastive Estimation. Lastly, we discuss related work to CPC.

2.1 Motivation and Intuitions

The main intuition behind our model is to learn the representations that encode the underlying shared information between different parts of the (high-dimensional) signal. At the same time it discards low-level information and noise that is more local. In time series and high-dimensional modeling, approaches that use next step prediction exploit the local smoothness of the signal. When predicting further in the future, the amount of shared information becomes much lower, and the model needs to infer more global structure. These 'slow features' [13] that span many time steps are often more interesting (e.g., phonemes and intonation in speech, objects in images, or the story line in books.).

One of the challenges of predicting high-dimensional data is that unimodal losses such as mean-squared error and cross-entropy are not very useful, and powerful conditional generative models which need to reconstruct every detail in the data are usually required. But these models are computationally intense, and waste capacity at modeling the complex relationships in the data x, often ignoring the context c. For example, images may contain thousands of bits of information while the high-level latent variables such as the class label contain much less information (10 bits for 1,024 categories). This suggests that modeling $p(x|c)$ directly may not be optimal for the purpose of extracting shared information between x and c. When predicting future information we instead encode the target x (future) and context c (present) into a compact distributed vector representations (via non-linear

Figure 1: Figure 1 (paper): overview of Contrastive Predictive Coding (CPC).